

Nathan Hammerschmitt Le Gal

nathan.p.hammerschmitt.le.gal.27@dartmouth.edu | (781) 584-2772 | linkedin.com/in/nathan-hammerschmitt-le-gal

Education

Dartmouth College, Hanover, NH

GPA: 3.83/4.0

Bachelor of Arts: Engineering Sciences, Studio Art Modifier

Jun 2027

Bachelor of Engineering: Mechanical Engineering

Jun 2028

Selected Coursework: Solid Mechanics, Materials Science, Kinematics & CAD, Embedded Systems, Digital Circuits, Physics, Differential Equations, Sculpture, Drawing

Skills

CAD & Design: SolidWorks (CSWA certified), Fusion 360, Blender, GD&T, tolerance analysis, technical drawings

Programming: Python, C, Matlab, Arduino, Git, data processing, scripting & automation

Manufacturing & Prototyping: FDM 3D-printing, laser cutting, CAM, DFM & DFA, iterative design

Analysis: FEA (SolidWorks), SAP2000, kinematic modeling, material characterization, experimental validation

Work Experience

Nexus Footwear Company, *Mechanical Engineer & Product Lead*

Hanover, NH | **Jan 2025 – Present**

- Designed a custom-fit, 3D-printed shoe using scan data, focusing on mechanical fit and manufacturability
- Produced 50+ design-test iterations, refining geometry, tolerances, and print parameters to meet fit, durability, and surface-quality targets
- Scaled production to 80 units/month by building a 6-printer workflow with automated g-code generation
- Secured \$40k in R&D funding by defining technical milestones, validation goals, and deliverables

UMass Amherst Integrated Locomotion Lab, *Research Assistant*

Amherst, MA | **Jun 2024 – Aug 2024**

- Analyzed accelerometer data with 30+ python scripts to derive biomechanical metrics of outdoor running
- Assisted graduate students to facilitate motion-capture, IMU, and metabolic data collections

Projects

Portfolio: <https://nathanhlg.com/>

Truss Bridge, Solid Mechanics

Aug 2025

- Led 3-person team to build plywood bridge (0.84 kg) that carried 20 kN during Instron compression test
- Used Solidworks FEA and SAP2000 to iterate on truss geometry to maximize strength to weight ratio
- Produced 2D drawings and fabrication plans, and correlated compression test results with simulations

Dino Robot Walker, Computer Aided Design and Kinematics

May 2025

- Designed bipedal robot using a rotational 4-bar linkage modeled in GeoGebra
- Created drawings and documentation of the mechanical system for repeatable fabrication and assembly

Leadership & Community Engagement

Dartmouth Triathlon & Cycling Teams

Competitive Athlete & Equipment Executive

Hanover, NH | **Sep 2023 – Present**

- Raced 2025 Triathlon Collegiate Club Nationals, 7 sprint triathlons, 3 half marathons
- Manage \$25k of team equipment for 40 active team members
- Design and order Castelli performance kits, coordinating supplier specs, sizing, and delivery

Additional Information

Accomplishments: 4 Academic Citations (awarded to 2.4% of students), Leapyear Startup Accelerator (\$30k), Henry David Thoreau Scholarship (\$25k), Cell Signaling Technology Scholarship (\$10k), Harry Agganis Scholarship (\$8k), High School Valedictorian

Languages: English, French

Interests: Additive Manufacturing, Automation, Robotics, Product Design, Wearables, Entrepreneurship, Industrial Design